Syllabus for

Academic year 2020/2021 >

FFM521 - Mechanics 2

Mekanik 2

Syllabus adopted 2019-02-14 by Head of Programme (or corresponding)
Owner: TKTFY
6,0 Credits
Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail
Education cycle: First-cycle
Major subject: Engineering Physics
Department: 16 - PHYSICS

Teaching language: Swedish
Application code: 57145
Open for exchange students: No
Maximum participants: 130
Only students with the course round in the programme plan

Credit distribution

Module				Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates		
0112	Examination	4,5 c	Grading: TH				4,5 c			Thu 03/06-2021 am J,	Fri 09/10-2020 am J,	Mon 16/08-2021 am J
0212	Project	1,5 c	Grading: UG				1,5 c					

In programs
TKTFY ENGINEERING PHYSICS, Year 1 (compulsory)

Examiner:
Martin Cederwall

Go to course homepage

Eligibility
General entry requirements for bachelor's level (first cycle)
Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements
The same as for the programme that owns the course.
Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites
Tools of Engineering Physics, Linear algebra and geometry, Introductory mathematical analysis, Real analysis and Mechanics 1 or corresponding course.

Aim
- To provide a good understanding of the basic concepts of mechanics, that provides a basis for further physics studies.
- To provide training in the translation of a physics problem into a mathematical model, and to analyse this model.

Learning outcomes *(after completion of the course the student should be able to)*

Kinematics and dynamics of rigid bodies - the student should use their understanding of particle systems to be able to derive and apply the laws of rotational and translational motion.
Motion in non-inertial systems - The student will gain an understanding of fictitious forces, in particular centrifugal and Coriolis forces in the rotating reference system.

The theory of harmonic motion - the student should be able to apply their knowledge of mathematics to the damped and undamped, free as well as forced harmonic motion.

Fundamental principles; Symmetries, conservation laws. Galilean invariance and the relativity principle.

Analytical Mechanics - The student should have been offered an initial acquaintance with analytical mechanics. The student will be able to study simple particle system using the Lagrange formalism.

Develop communication - Being able to present the basic mechanical concept in writing.
Overall - Knowledge of all the above areas shall be used in problem solving. The student should be able to identify the relevant information in a mechanics and then formulate the problem mathematically using tools from mechanics and mathematics.

Content

Plane kinematics of rigid bodies
Plane kinetics of rigid bodies
Rotating systems and inertial forces
Three-dimensional kinematics and dynamics of rigid bodies
Vibration and time response

Symmetries, conservation laws and the relativity principle

Analytical mechanics

Organisation

Lectures, problem solving sessions, web-based practice problems together with few obligatory problems taking the form of miniprojects.

Literature

Announced on the course web page.

Examination including compulsory elements

Written exam and a project consisting of obligatory homework problems.

In order to pass the project part of the course, it is necessary to pass these and to obtain a pass on a similar problem which is part of the written exam. During the written examination at most one hour can be spent to solve the project related problem.

The written exam consists of six problems, one of them examining only the project. The remaining five problems are the foundation for the mark on the whole course.